

Movie Revenue Prediction: A Comparative Study of Six Regression Algorithms

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June 2026

ABSTRACT

Predicting global box-office revenue from pre-release metadata is a critical yet challenging task due to the high volatility and extreme right-skewness of movie returns. Using the canonical TMDb Box Office dataset of 3,000 films, we engineer a comprehensive 49-feature space spanning budget dynamics, popularity indicators, temporal release windows, cast gender compositions, and metadata completeness signals. To normalize a target distribution spanning nine orders of magnitude, we implement a $\log(1+y)$ target transformation and benchmark six regression architectures: Linear Regression, Ridge Regression, Random Forest, XGBoost, LightGBM, and CatBoost. Evaluated under a rigorous 5-fold cross-validation pipeline, the gradient-boosted tree architectures significantly outperform linear baselines. CatBoost achieves the highest predictive stability with a cross-validated RMSLE of 1.584 ± 0.029 , while Random Forest demonstrates a hold-out R^2 of 0.709 and a Median Absolute Percentage Error (MedAPE) of 66.5%. A formal ablation study reveals that budget profiles, popularity metrics, and temporal configurations serve as the primary drivers of model performance.

1. Introduction

Evaluating the financial viability of a cinematic project prior to its release involves navigating complex, non-linear interactions between budget constraints, cultural trends, and temporal scheduling. In data science, this is framed as a supervised regression problem characterized by heterogeneous feature types, a modest sample size, and a highly heavy-tailed target variable. Predictive modeling on raw revenue scales frequently misbehaves due to the presence of low-budget independent features alongside multi-million-dollar blockbusters.

This study develops a robust, reproducible machine learning framework to systematically predict movie revenue using metadata extracted from The Movie Database (TMDb). We move beyond simple linear assumptions by mapping out a 49-dimensional feature space and evaluating six diverse machine learning models representing both linear equations and tree-based ensembles. Our methodology optimizes performance metrics for heavy-tailed targets, introduces robust error evaluation to circumvent standard statistical traps, and performs an algorithmic ablation analysis to break down the exact performance contributions of distinct feature sub-groups.

2. Dataset and Feature Engineering

The source dataset comprises 3,000 labeled records containing detailed metadata fields such as production budgets, language settings, release schedules, and complex JSON-encoded metadata structures documenting cast, crew, production companies, and keywords. Initial data auditing revealed structural placeholders where unknown revenues were filled in with values $\leq \$1,000$. These entries represent non-random data noise and were removed during data cleaning, resulting in a core dataset of 2,943 usable records.

The raw box-office revenue distribution exhibits extreme right-skewness, where a small cohort of tentpole releases commands billions in revenue while the median release centers around \$17 million. Modeling this distribution directly under standard least-squares objectives introduces severe instability. Consequently, we apply a log target transformation, $\mathbf{z} = \log(\mathbf{1} + \mathbf{y})$, which aligns the distribution with approximately Gaussian characteristics, stabilizes variance, and ensures that standard loss functions evaluate relative variations equitably across different scales of bütçe and success.

To capture latent predictive signals within the raw columns, we parse the text and JSON structures to construct 49 distinct numerical, binary, and categorical variables. These are organized into eight functional groups as detailed in Table 1.

Feature Group	# Features	Core Dimensions & Examples
Budget Profiles	4	$\log(1+\text{budget})$, budget_per_minute , missing indicator flag
Popularity Metrics	2	Continuous popularity trend score, absolute runtime
Temporal Context	7	Release year, month, decade, day of week, summer/holiday season flags
Structural Counts	10	Total cast size, total crew size, male/female/unknown cast counts
Presence Flags	6	Franchise membership (has_collection), homepage existence, title translations
Text Metadata	4	Character lengths of overview summary, title string, and tagline
Genre Indicators	15	One-hot binary flags mapping membership to the top 15 genres
Language Controls	1	Categorical tracking for top 10 primary languages + grouped "other"
Total Space	49	<i>Engineered predictive matrix</i>

Table 1: Structured categorization of the 49 engineered metadata features.

3. Methodology

3.1 Model Architectures Under Review

To ensure a fair evaluation across model families, we test six distinct regression architectures:

- **Linear Regression:** Fits a standard least-squares hyperplane to establish a traditional baseline.
- **Ridge Regression:** Introduces an L_2 regularization penalty ($\alpha = 1.0$) to control parameter coefficients and prevent instability from collinear features.
- **Random Forest:** An ensemble bagging model that fits 400 de-correlated decision trees to smooth variance ($\text{max_depth} = 12$).
- **XGBoost:** A gradient boosting framework utilizing second-order Taylor expansion loss approximations alongside structural tree pruning (600 estimators, $\text{lr} = 0.05$).
- **LightGBM:** Optimizes training velocity by growing decision trees leaf-wise rather than level-wise and employing histogram-based feature binning (800 estimators, 31 leaves).
- **CatBoost:** Uses symmetric, oblivious decision trees paired with an ordered boosting mechanism designed to counter target leakage and overfitting on smaller training sets (800 iterations).

3.2 Training Pipeline and Evaluation Metrics

All data processing is orchestrated via a synchronized scikit-learn pipeline, ensuring that feature imputation (median value tracking for continuous variables; mode tracking for categorical sets) and one-hot encodings are computed exclusively within individual training folds. This strict isolation prevents data leakage during validation.

Models are evaluated using standard metrics back-transformed into real dollar spaces. In addition to standard Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2), we track Root Mean Squared Log Error (RMSLE), Mean Absolute Percentage Error (MAPE), and Median Absolute Percentage Error (MedAPE).

A crucial statistical consideration is the unreliability of MAPE on heavy-tailed distributions. Because MAPE normalizes error against the true target value, low-budget films that generate modest deviations in absolute dollars produce massive percentage distortions. For instance, predicting a \$1.5 million return for a project that earned \$40,000 creates a local error exceeding 3,600%, heavily biasing the overall average error. To balance this distortion, we designate RMSLE as our primary benchmark and introduce MedAPE as a robust alternative that tracks the typical relative prediction error without being skewed by a few extreme outliers.

4. Empirical Results

Headline performance metrics across a standardized 80/20 train/test hold-out partition and a 5-fold cross-validation routine are consolidated in Table 2.

Model Architecture	RMSE (\$M)	R^2	RMSLE	MedAPE	MAPE	CV-RMSLE
Linear Regression	128.1	0.199	1.669	69.8%	966%	1.777 $\pm$ 0.066
Ridge Regression	129.5	0.182	1.670	69.4%	968%	1.776 $\pm$ 0.065
Random Forest	77.2	0.709	1.577	66.5%	574%	1.614 $\pm$ 0.032
XGBoost	77.3	0.709	1.569	65.5%	612%	1.614 $\pm$ 0.044
LightGBM	83.9	0.656	1.582	65.3%	547%	1.642 $\pm$ 0.048
CatBoost	82.3	0.670	1.539	65.4%	561%	1.584 $\pm$ 0.029

Table 2: Performance summary across models. RMSE values are denoted in millions of dollars. Bold markers isolate top-performing metrics.

The tree-based ensemble family outperforms the linear baselines by a large margin across all metrics. Linear frameworks struggle with severe under-prediction at the higher ends of the revenue scale and suffer from high heteroscedasticity. By contrast, tree ensembles successfully map the full range of revenue without catastrophic failures. CatBoost records the lowest cross-validation error (**1.584**) and the highest stability (lowest standard deviation), while Random Forest and XGBoost achieve peak hold-out variance explanation with an R^2 of 0.709.

4.1 Ablation Diagnostics

To quantify the exact predictive value of the engineered feature domains, we execute a leave-one-group-out cross-validation ablation sequence using our top-performing CatBoost architecture. The results are tracked via the delta shift in cross-validated error ($\Delta RMSLE = RMSLE_{ablated} - RMSLE_{baseline}$), as shown in Table 3.

Feature Group Omitted	Resulting CV-RMSLE	Δ RMSLE vs. Baseline
Full Baseline Framework	1.584	Reference
- Budget Profiles	1.717	+0.133
- Popularity Metrics	1.693	+0.110
- Temporal Context	1.665	+0.081
- Genre Indicators	1.613	+0.029
- Presence Flags	1.593	+0.009
- Language Controls	1.591	+0.007
- Structural Counts	1.585	+0.001
- Text Metadata	1.568	-0.016

Table 3: Multi-group feature ablation impact on CatBoost cross-validation error.

The ablation sequence highlights three primary predictive engines: bütçe profiles (+0.133), popularity metrics (+0.110), and temporal context features (+0.081). Genre configurations provide a minor but clear positive signal. Conversely, raw text-length variables slightly degrade generalization, acting as noise for this sample size; removing them improves the cross-validated RMSLE to 1.568.

5. Discussion and Limitations

The clear superiority of tree ensembles stems from their capacity to map complex, conditional interactions that linear equations miss. For example, a linear model assumes a fixed relationship between bütçe allocation and financial returns. Tree ensembles, however, easily capture non-linear realities, such as the fact that bütçe increases yield higher relative returns for action franchises than for dramatic features.

The stable MedAPE of approximately 65% confirms that the tree ensembles deliver highly practical predictive utility. Given that the input features are restricted to pre-release technical metadata and lack downstream variables like active marketing expenditure, critic scores, or opening weekend theater counts, predicting revenue within a 65% median boundary demonstrates strong functional performance. The remaining mass of the MAPE distribution is driven by lower-tier independent records where small absolute dollar errors appear as massive percentage distortions due to the small denominator.

A primary limitation of this work is the sample size of 3,000 films, which restricts our ability to leverage deep entity embeddings for specific cast and crew identities. Additionally, box-office receipts are evaluated in nominal dollars across historical decades; integrating inflation-adjusted targets would likely sharpen the signal of temporal and historical bütçe features.

6. Conclusion

This study demonstrates a robust machine learning system for movie revenue prediction using the TMDB dataset. By utilizing a log-target transformation and engineering a 49-dimensional feature space, tree-based ensemble methods successfully capture the underlying non-linear distribution of box-office returns. CatBoost achieves the highest cross-validation stability with an RMSLE of 1.584 ± 0.029 , while Random Forest establishes a high hold-out baseline with an R^2 of 0.709 and a MedAPE of 66.5%. Feature ablation confirms that bütçe structures, public popularity tracking, and seasonal temporal scheduling serve as the foundational drivers of box-office prediction performance.

References

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